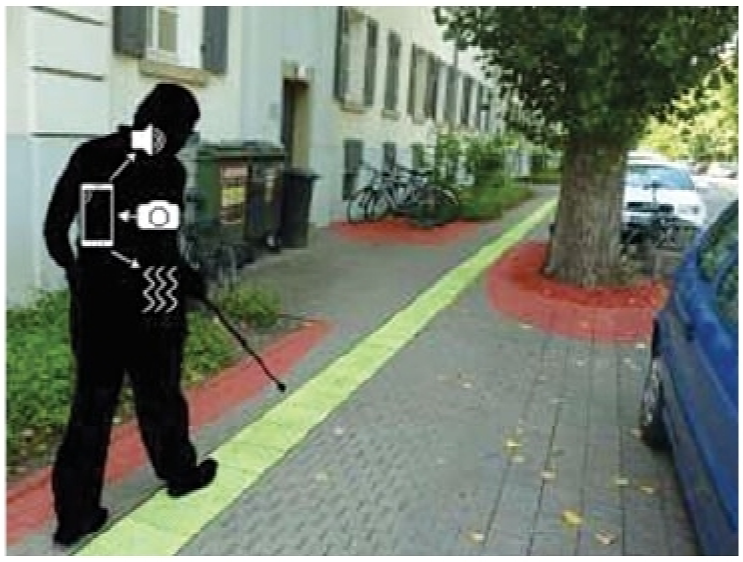

*A mobile-based navigational aid for visually-impaired persons (Credit: https://phys.org)*

head motion and to a belt at the waist to monitor body balance disorders. A handheld control provides real-time visual and/or auditory feedback to the user and communicates patient status to a remote clinician.

*Wearable device for healing fractures.* The device is used for accelerating the healing process of long bones for patients with open fractures. A pair of miniature ultrasound transducers is implanted into the affected region and is wirelessly connected to a unit located in a clinical environment. The device initiates low-intensity pulsed ultrasound therapy sessions, carries out ultrasound measurements and communicates the data to a centralised unit.

## The future

Basic wearable medical devices such as headphones, hearing aids, health-bands help improve the quality of life. Functional aids used by mentally- or physically-disabled and chronically-ill persons provide higher levels of independence, and enable their social and vocational participation.

Technological advancements have made available wider data memory capability, augmented reality (AR)-based head-mounted displays, AI-based decision-making capability and improved communication. Advanced and superior components and sensors are helping in improving functionality, wearability, reliability and security of devices.

Devices are being integrated with implantable devices to provide real-time drug delivery, functional stimulation of the nervous system and heart management. Futuristic wearable devices will include automatic infusion of insulin when blood sugar levels are high, serve as implanted micro-pump assisted drug delivery systems, DNA diagnostics implants, artificial retinas, and disposable sensors for monitoring glucose, urea, sodium, and calcium.

An advanced GPS-based wearable navigation and guidance system for visually-impaired patients is under development to assist them in moving to the desired destination. The system includes a GPS receiver to locate the patient in a spatial GIS database, a compass for determining orientation, a cellphone for communicating with a central server, and a speech synthesiser to make available to the user information about nearby streets and other objects of interest.

Stringent requirements for clinical testing, applicable legislation and regulations, and potential threats of litigation are some of the limitations responsible for the long period taken from conceptualisation to commercial availability of wearable devices. Wearable medical devices are treated more as complementary rather than an alternative to main hospital equipment. However, their ever-increasing capability combined with their practical benefits in providing unconfined medical monitoring, assistance in rehabilitation, improvement in the living conditions of chronically-ill patients and point-of-care service in remote locations make them highly useful.

These devices have a distinct advantage in providing telemedicine services, data transfer security and easy integration. These are making available vast useful information and data for research purposes, and providing effective health services in emergency situations.

Demand for medical devices is growing at a fast rate as people become more and more health conscious with increasing government-aided social healthcare schemes and an increasing number of patients moving from hospitals to their homes for nursing and rehabilitation. EFY